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Comparisons of classical and Wigner sampling of transition state energy levels for quasiclassical trajectory chemical dynamics simulations

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Quasiclassical trajectory calculations are compared, with classical and Wigner sampling of transition state (TS) energy levels, for $\text{C}_2\text{H}_5\text{F}^\ddagger \rightarrow \text{HF} + \text{C}_2\text{H}_4$ product energy partitioning and $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier dynamics. The calculations with Wigner sampling are reported here for comparison with the previously reported calculations with classical sampling [Y. J. Cho *et al.*, J. Chem. Phys. **96**, 8275 (1992); L. Sun and W. L. Hase, J. Chem. Phys. **121**, 8831 (2004)]. The $\text{C}_2\text{H}_5\text{F}^\ddagger$ calculations were performed with direct dynamics at the MP2/6-31G* level of theory. Classical and Wigner sampling give post-transition state dynamics, for these two chemical systems, which are the same within statistical uncertainties. This is a result of important equivalences in these two sampling methods for selecting initial conditions at a TS. In contrast, classical and Wigner sampling often give different photodissociation dynamics [R. Schinke, J. Phys. Chem. **92**, 3195 (1988)]. Here the sampling is performed for a vibrational state of the ground electronic state potential energy surface (PES), which is then projected onto the excited electronic state's PES. Differences between the ground and the excited PESs may give rise to substantially different excitations of the vibrational and dissociative coordinates on the excited state PES by classical and Wigner sampling, resulting in different photodissociation dynamics. © 2010 American Institute of Physics. [doi:10.1063/1.3463717]

I. INTRODUCTION

Classical trajectory chemical dynamics simulations^{1–3} are widely used to study the atomic-level behavior of chemical reactions,⁴ intramolecular vibrational energy redistribution,^{5,6} and collisional energy transfer.⁷ Though the fundamental atomic-level motion is quantum mechanical,⁸ classical dynamics often gives near quantitative agreement with experiment for direct chemical processes that are classically allowed. These processes include direct bimolecular reactions,^{9,10} dynamics on repulsive potential energy curves,^{11,12} short-time unimolecular decomposition^{13–15} and intramolecular vibrational energy flow,^{5,6} and gas-surface energy transfer.^{16,17} Trajectories do not describe quantum effects such as tunneling through potential energy barriers, zero-point energy constraints, and transitions between potential energy surfaces for different electronic states.¹⁸ For the latter, a semiclassical algorithm may be used to include electronic transitions in classical trajectory calculations.¹⁹ Unphysical zero-point energy flow becomes particularly important for long-time nondirect trajectories.²⁰

To obtain accurate results from a chemical dynamics simulation, it is paramount that proper initial conditions (i.e., coordinates and momenta) are chosen for the ensemble of trajectories propagated to compare with experiment. The quasiclassical sampling method is widely used to select initial conditions for classical trajectories.^{2,3} For this method,

initial reactant energies are chosen to match those for the reactant's quantum mechanical energy levels. These energies are then transformed to the coordinates and momenta for the classical trajectory calculations. For example, to transform a normal mode vibrational energy to its coordinate and conjugate momentum, a random phase is often chosen for the normal mode. The resulting distribution is a classical one with the most probable values for the normal mode coordinate being at the classical turning points. As shown in Fig. 1, for vibrational state $n=0$, the classical distribution is decidedly different from the quantum one, which is peaked at the equilibrium value. For large n , the quantum and classical distributions become similar.

To sample a quantum mechanical distribution of an initial state in phase space, the obvious problem encountered is that the wave function is a function of either coordinates or momenta, while the phase space is a $2N$ -dimensional space spanned by both coordinates and their conjugate momenta. Defining a quantum mechanical distribution determined simultaneously by the coordinates and momenta in classical phase space, which at the same time mimics as closely as possible the quantum mechanical distribution, is an old theme in theoretical physics.²¹ A commonly used quantum distribution function in classical phase space, and the one considered here, is the Wigner distribution function.^{22–24} Other types of distribution functions have also been used.^{25–30}

The Wigner distribution function^{22–24} may be used to select initial coordinates and momenta for trajectory calcula-

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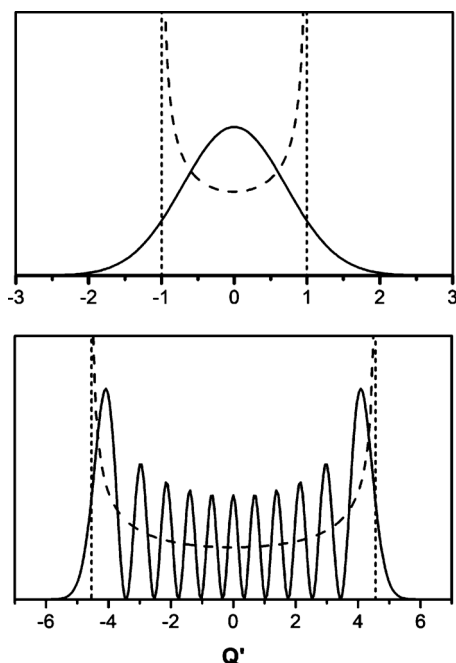


FIG. 1. Comparison of quantum and classical probability distributions of the normal mode coordinate $Q' = \sqrt{\omega/\hbar}Q$ for the quantum states $n=0$ and 10 .

tions of unimolecular processes. For bimolecular trajectory calculations, the classical random phase should be used so that the trajectory results do not depend on the initial reactant separation.³¹ A number of different studies have been performed with Wigner distribution function including semiclassical dynamics and the classical-quantum correspondence.^{24,32} trajectory calculations of unimolecular dissociation^{23,33,34} and photodissociation,^{35–41} and representation of initial and final states in classical trajectories.^{42–44} The importance of choosing trajectory initial conditions, using the Wigner distribution, has been emphasized for molecular photodissociations.^{23,25,28,37,45–49} However, for vibrational states with large n , the classical and Wigner distributions give similar initial coordinates and momenta and their ensuing dynamics are similar.²⁸ In addition, even at low n , where the classical and Wigner distributions are decidedly different, they may give similar photodissociation dynamics.⁴⁵

In recent research, classical trajectory simulations have been used to study post-transition state (TS) dynamics for unimolecular dissociation reactions.⁵⁰ Initial coordinates for the trajectories were chosen at the dissociation TS in accord with microcanonical and canonical ensembles,^{3,51,52} as well as for specific vibrational/rational states.⁵³ Post-transition state dynamics, with its potential energy release and motion from a TS to products, is similar to photodissociation dynamics on a repulsive potential energy surface (PES). For these post-transition state dynamics studies, classical sampling was used to transform the initial condition's vibrational states to coordinates and momenta. Of interest for the work presented here is whether the calculated post-transition state dynamics would differ if instead Wigner sampling was used for transforming the vibrational states to coordinates and momenta. Such a comparison between classical and Wigner sampling is made for $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier dynamics and

$\text{C}_2\text{H}_5\text{F}^\ddagger \rightarrow \text{HF} + \text{C}_2\text{H}_4$ product energy partitioning, for which the post-transition state dynamics with classical sampling of the TS energy levels has been reported previously.^{51,54–57}

The remainder of this article is organized as follows. Section II reviews the Wigner distribution function and its properties and describes how to sample it for quasiclassical trajectory simulations. Comparisons of simulation results for $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier dynamics and $\text{C}_2\text{H}_5\text{F}^\ddagger \rightarrow \text{HF} + \text{C}_2\text{H}_4$ product energy partitioning for the initial conditions chosen by sampling classical and Wigner coordinate and momenta distributions are given in Sec. III. The calculations for the Wigner distribution samplings are new and were performed for this study. The article concludes with a discussion in Sec. IV.

II. WIGNER DISTRIBUTION FUNCTION

A. Functional form and properties

The Wigner distribution (i.e., density) function $W(\mathbf{q}, \mathbf{p})$ gives the classical probability density, which reproduces the quantum distributions in positions $\mathbf{q}$ and momenta $\mathbf{p}$. Given a quantum mechanical wave function $\Psi(\mathbf{q})$, the Wigner distribution function is defined as²²

$$W(\mathbf{q}, \mathbf{p}) \equiv \frac{1}{(2\pi\hbar)^N} \int d\mathbf{s} \exp(i\mathbf{p} \cdot \mathbf{s}/\hbar) \Psi(\mathbf{q} - \mathbf{s}/2) \Psi(\mathbf{q} + \mathbf{s}/2), \quad (1)$$

where $\mathbf{s}$ is a spatial variable and N is the number of dimensions. This function satisfies the normalization condition

$$\int \int W(\mathbf{q}, \mathbf{p}) d\mathbf{q} d\mathbf{p} = 1 \quad (2)$$

and when integrated over the momenta yields the quantum mechanical coordinate probability distribution $|\Psi(\mathbf{q})|^2$, i.e.,

$$\int W(\mathbf{q}, \mathbf{p}) d\mathbf{p} = |\Psi(\mathbf{q})|^2. \quad (3)$$

The problem of interest here is sampling the Wigner distribution for the coordinates $\mathbf{Q}$ and momenta $\mathbf{P}$ of N uncoupled harmonic oscillators, for which the wave function is a product of one-dimensional harmonic oscillator wave functions; $\Psi(\mathbf{Q}, \mathbf{P}) = \prod_{i=1}^N \varphi(Q_i)$. The corresponding Wigner distribution is

$$W(\mathbf{Q}, \mathbf{P}) = \prod_{i=1}^N W(Q_i, P_i). \quad (4)$$

To consider the form of $\Psi(\mathbf{Q}, \mathbf{P})$, define the dimensionless coordinate and momentum for a single harmonic oscillator

$$\begin{aligned} Q' &= \sqrt{\omega/\hbar} Q, \\ P' &= P/\sqrt{\omega\hbar}, \end{aligned} \quad (5)$$

where $\omega = 2\pi\nu$ and ν is the vibrational frequency. The wave function for a harmonic oscillator in its n th vibrational state is

$$\varphi_n = 2^{-n/2} (n!)^{-1/2} \pi^{-1/4} \exp\left(-\frac{1}{2}Q'^2\right) H_n(Q'), \quad (6)$$

where H_n is the n th Hermite polynomial. If this wave function is inserted into Eq. (1), the Wigner distribution for each Q_i, P_i pair in Eq. (4) takes the form

$$W^{(n)}(Q', P') = (-1)^n (\pi n!) L_n(\rho) e^{-\rho^2/2}, \quad (7)$$

where L_n is the n th order Laguerre polynomial and ρ is given by

$$\rho^2 = 2(P'^2 + Q'^2). \quad (8)$$

For the vibrational ground state, $n=0$, $W^{(n=0)}(Q, P)$ in Eq. (7) is a simple form and the product of two Gaussian functions

$$W^{(n=0)}(Q', P') = \frac{1}{\pi \hbar} e^{-Q'^2} e^{-P'^2}. \quad (9)$$

From the definition of Q' and P' in Eq. (5), Eq. (9) may also be written as

$$W^{(n=0)}(Q, P) = \frac{1}{\pi \hbar} e^{-(P^2 + \omega^2 Q^2)/\omega \hbar}. \quad (10)$$

Since the calculated energy for the harmonic oscillator is $E = (P^2 + \omega^2 Q^2)/2$, Eqs. (9) and (10) may be expressed as

$$W^{(n=0)}(E) = \frac{1}{\pi \hbar \nu} e^{-2E/\omega \hbar}, \quad (11)$$

where Hamilton's equation have been used to equate $dQ' dP'$ to $dEdt$. Integrating dt over one period yields $1/\nu$, resulting in Eq. (11). The average value of E is

$$\langle E \rangle = \int_0^\infty E W^{(n=0)} dE = \hbar \nu / 2, \quad (12)$$

which is the energy of the $n=0$ level. In general, $E = (n+1/2)\hbar\nu$.

From Eq. (9), it is seen that the Wigner (i.e., quantum mechanical) distribution for both Q' and P' , and thus Q and P , peaks and is centered at zero for $n=0$. The distribution for Q is totally different from that for a classical harmonic oscillator, for which the most probable value for Q is found at the oscillator's classical turning points with $P=0$. The difference between the classical and the quantum distributions of Q decreases with an increase in n . The quantum and classical distributions for Q are compared in Fig. 1 for n equal to 0 and 10.

B. Sampling the Wigner distribution function

For a polyatomic molecule with N normal modes of vibration, there are two general approaches for sampling the Wigner distribution for the molecule's vibrational state specified by the molecule's N quantum numbers $\mathbf{n}$. For one, the total energy is not fixed and the Wigner distribution for each mode is sampled independently. For the second, an ensemble with a fixed total energy is sampled, properly weighted by the Wigner distribution for all N modes. Either sampling approach may be used for a simulation of a molecule in a specific quantum state $\mathbf{n}$. In addition, the samplings may be for simulations of microcanonical and canonical

ensembles.^{3,50,51} Here state $\mathbf{n}$ is chosen randomly for the ensemble and then sampled in accord with its Wigner distribution. Both the nonfixed and fixed energy sampling approaches are described in the following. A fixed energy sampling approach, based on an efficient orthant sampling algorithm, is used for the current study.

1. Nonfixed energy, independent mode sampling

For a single harmonic oscillator, Q and P are sampled independently as shown by Eq. (9) for the ground vibrational state. Though this sampling is performed without fixing the total energy, the resulting average energy sampled is that for the vibrational state being sampled, i.e., Eq. (12). If the energy for the oscillator was fixed, and either Q or P sampled by its Wigner distribution, the resulting distribution for the P or Q not sampled would not be that of Wigner.

For a molecule with N oscillators, each oscillator containing n quanta is sampled independently in accord with the Wigner distributions for its Q and P . The distributions of energies for the oscillators, which are independent, may be convoluted to obtain the distribution for the total energy of the molecule, i.e., for two oscillators the convolution is

$$P(E) = \int_0^E W_1(E_1) W_2(E - E_1) dE_1, \quad (13)$$

where W_1 and W_2 are the Wigner distributions for the two oscillators.

2. Fixed energy sampling

The sampling of fixed energy initial conditions for a polyatomic molecule, according to the Wigner distribution, can be achieved in various ways. One is to generate phase space points uniformly and weight the trajectory results with the Wigner distribution function.^{36,45} Another approach is to directly generate points in the classical phase space according to the Wigner distribution (importance sampling).^{23,25} From a computational point of view, the former approach is easier to implement since there are several ways to randomly select a phase space point.³ However, this approach requires a large number of trajectories to obtain the converged results. Since in the latter approach, phase space points are generated preferentially in accord with the Wigner distribution function, a much smaller number of trajectories are needed.

In the following, an efficient importance sampling approach is presented for sampling a fixed energy Wigner distribution for a polyatomic molecule. It is based on an efficient classical microcanonical sampling method.^{58,59} Classical microcanonical sampling for a system with N degrees of freedom is equivalent to choosing a uniform distribution of points in the $2N$ -dimensional molecular phase space bounded by hypersurfaces of constant energy H and $H+dH$.⁵⁸ The most straightforward approach is to select in a purely random fashion $2N-1$ of the coordinates and momenta in a hyper-rectangle box, bounded by values allowed by the total energy, with the remaining coordinate or momentum chosen by the total energy conservation. However, the remaining variable, for instance, P_N , is not sampled uni-

formly between H and $H+dH$. The proper weighting for selecting P_N is seen from the following transformation for the volume element $d\mathbf{Q}d\mathbf{P}$ in Eq. (14):⁵⁸

$$dQ_1 \cdots dQ_N dP_1 \cdots dP_N = dQ_1 \cdots dQ_N dP_1 \cdots dP_{N-1} dH \left| \frac{\partial H}{\partial P_N} \right|^{-1}. \quad (14)$$

(Here $\mathbf{Q}$ and $\mathbf{P}$ are used to represent general coordinates and momenta since $d\mathbf{Q}d\mathbf{P}$ is independent of the coordinate system.⁶⁰) The relative weighting factor for selecting P_N is then

$$P(P_N) \propto \left| \frac{\partial H}{\partial P_N} \right|^{-1}, \quad (15)$$

which according to Hamilton's equations is $|P_N|^{-1}$ for a normal mode Hamiltonian. The sampled points are then selected using rejection techniques, such as the von Neumann rejection technique.⁶¹ If a coordinate Q_N is chosen from energy conservation instead of a momentum, its relative probability will naturally be

$$P(Q_N) \propto \left| \frac{\partial H}{\partial Q_N} \right|^{-1}. \quad (16)$$

To extend the above method so that it may be used to sample the Wigner distribution for a polyatomic molecule with N normal modes of vibration, values for $Q_1, Q_2, \dots, Q_N$ and $P_1, P_2, \dots, P_{N-1}$ are selected randomly within a range of accessible values including values for the Q_i in their classically forbidden regions. P_N is then selected by energy conservation. The relative probability for the selected $Q_1, Q_2, \dots, Q_N$ and $P_1, P_2, \dots, P_N$ phase points is then

$$W(Q, P) |P_N|^{-1}, \quad (17)$$

where $W(Q, P)$ is given by Eq. (4). This probability may be sampled by the von Neumann rejection technique.⁶¹

3. Fixed energy sampling using orthant sampling

A more efficient way, to sample a fixed energy Wigner distribution for n normal modes, is based on orthant sampling⁵⁹ and is the approach used in the current study. Orthant sampling uses a $2N$ -dimensional random unit vector and is an exact classical microcanonical sampling method for harmonic oscillators.⁶² Sampling within the classically forbidden regions of phase space is automatically included. The procedure for sampling the trajectory initial coordinates and momenta in this manner is as follows.

- (1) Find the total energy for N normal modes

$$E = \sum \frac{1}{2} \hbar \omega_i (P_i'^2 + Q_i'^2) = \sum \left(n + \frac{1}{2} \right) \hbar \omega_i. \quad (18)$$

- (2) Select a random point on the energy shell using a $2N$ -dimensional random unit vector.⁵⁹
- (3) The selected phase space point is then used to evaluate the Wigner distribution function, i.e., Eq. (4). von Neumann's rejection method⁶¹ is then employed in accepting or rejecting the point, i.e., accept the point if

$W/W_0 > R_i$, where W_0 is the most probable value of W and R_i is a fresh random number, otherwise go to step (2).

- (4) The accepted Q' and P' are then transformed to Cartesian coordinates in the same manner as for classical sampling, which has been described in detail previously.³

III. COMPARISON OF POST-TRANSITION STATE DYNAMICS FOR CLASSICAL AND WIGNER SAMPLING OF TRANSITION STATE ENERGY LEVELS

In this section the dynamics of ensembles of trajectories chosen by classical and Wigner sampling of energy levels are compared. The trajectories are initialized at the TS for both $C_2H_5F^\ddagger \rightarrow HF + C_2H_4$ dissociation and the $Cl^- + CH_3Cl \rightarrow ClCH_3 + Cl^-$ S_N2 nucleophilic substitution reaction. For the former reaction, interest is in the product energy partitioning to HF and C_2H_4 vibration and rotation and to the relative translation. The nature of the $[Cl \cdots CH_3 \cdots Cl]^\ddagger$ central barrier dynamics is the topic of interest for the latter reaction. There is extensive trapping of trajectories at the central barrier with a significant amount of central barrier recrossing. The question addressed here is whether classical and Wigner sampling of the TS energy levels give similar or different product energy partitioning for $C_2H_5F^\ddagger \rightarrow HF + C_2H_4$ dissociation and central barrier dynamics for $[Cl \cdots CH_3 \cdots Cl]^\ddagger$. Differences might be expected since the classical and Wigner sampling algorithms initialize the trajectories in different regions of phase space.

The simulations with classical sampling for $C_2H_5F^\ddagger \rightarrow HF + C_2H_4$ dissociation and $[Cl \cdots CH_3 \cdots Cl]^\ddagger$ central barrier dynamics have been reported previously.^{51,54-57} The former study is a comparison with experiments by Setser and co-workers.⁶³ Central barrier dynamics is an important component of the nonstatistical behavior of gas-phase S_N2 nucleophilic substitution reactions.⁶⁴

A. $C_2H_5F^\ddagger \rightarrow HF + C_2H_4$ product energy partitioning

1. Direct dynamics simulations with Wigner sampling

Direct dynamics simulations^{65,66} at the MP2/6-31G* level of theory were performed with Wigner sampling to compare with the previously reported⁵⁵ same type of direct dynamics, but performed with classical sampling of the TS energy levels. The simulations were performed with the chemical dynamics program VENUS (Refs. 67 and 68) interfaced with the GAUSSIAN (Ref. 69) electronic structure theory computer program. The trajectories were integrated with a Hessian-based predictor-corrector algorithm,⁷⁰ with the energies, gradient, and the Hessian calculated at MP2/6-31G* level of theory. The Hessian is updated five times before being recalculated.⁷¹ The integration step size is 0.25 amu^{1/2} bohr. The trajectories were stopped if the gradient between the fragments is less than 10^{-7} hartree/bohr or the distance between the fragments' center-of-mass is greater than 5 Å.

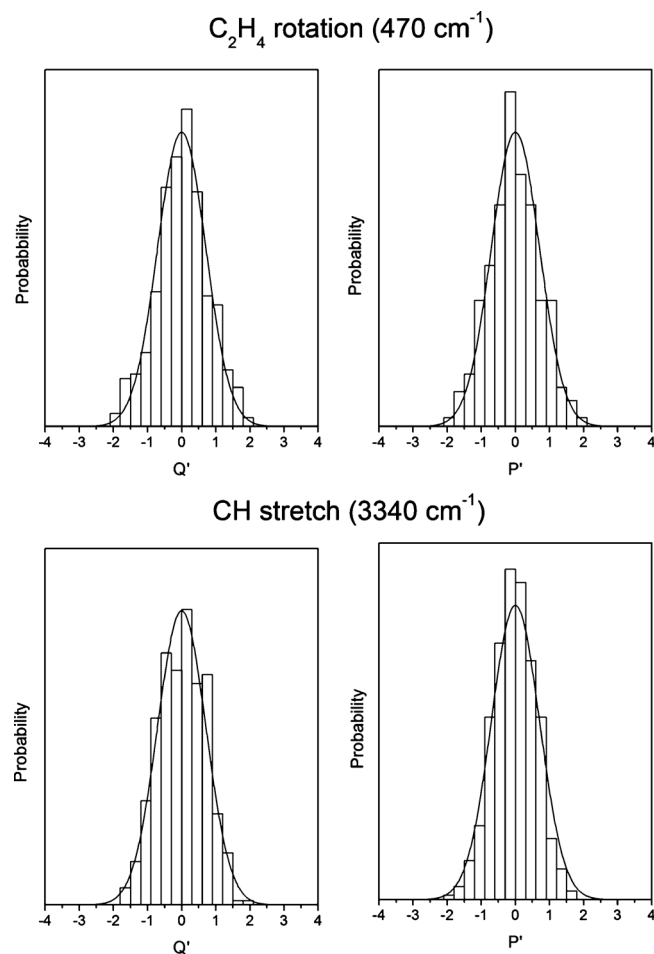


FIG. 2. Histograms comparing the actual Wigner distributions with the normal mode coordinates and momenta chosen by samplings of the Wigner distributions for the $\text{C}_2\text{H}_5\text{F}^\ddagger$ TS. Top figures are for the 470 cm^{-1} C_2H_4 rotation mode and the bottom figures for the 3340 cm^{-1} CH stretch mode.

The trajectories are initialized at the $\text{C}_2\text{H}_5\text{F}^\ddagger$ TS and directed toward the $\text{HF}+\text{C}_2\text{H}_4$ products. The normal mode/rigid rotor Hamiltonian is assumed for sampling the initial conditions, which are transformed to Cartesian coordinates and momenta used for calculating the classical trajectories.³ The reaction coordinate translational energy is 3.45 kcal/mol. Rotational energy is sampled from a 300 K Boltzmann distribution. Zero-point energy is added to the $3N-7=17$ vibrational modes at the TS. A total of 400 trajectories was calculated.

As illustrated in Fig. 2, the values for the normal mode coordinates and momenta, chosen for the trajectory initial conditions, are in very good agreement with the Wigner distributions. Thus, the above algorithm accurately samples the Wigner distribution.

2. Comparison of results for classical and Wigner sampling

The average percentages for the product energy partitioning, determined for the classical and Wigner sampling of the $\text{C}_2\text{H}_5\text{F}^\ddagger$ zero-point energy level, are compared in Table I. Overall, the average product energy partitionings for the two sampling methods are very similar and there are no statistical differences in the average percentage partitionings for any of

TABLE I. Product energy partitioning for $\text{C}_2\text{H}_5\text{F}^\ddagger \rightarrow \text{HF}+\text{C}_2\text{H}_4$. (The trajectory initial conditions are for zero-point vibrational energy, 3.45 kcal/mol in reaction coordinate translation, and rotational energy sampled from its 300 K Boltzmann distribution.)

Product energy	TS energy level sampling ^a	
	Classical ^b	Wigner
Rel. translation	75.4 ± 1.0	76.8 ± 1.0
C_2H_4 vibration	6.1 ± 1.6	4.8 ± 1.3
C_2H_4 rotation	4.6 ± 1.8	4.8 ± 1.8
HF vibration	10.5 ± 1.8	10.0 ± 1.9
HF rotation	3.4 ± 1.2	3.6 ± 1.1

^aThe average product energy partitioning percentages are listed. Zero-point energy is not included in the percentage for the C_2H_4 and HF vibration energies. The uncertainties are for the standard deviation of the mean.

^bThe results for the classical sampling of the TS energy levels were published previously in Ref. 54.

the product energies. Given the uncertainties in the standard deviation of the mean, classical and Wigner sampling of the $\text{C}_2\text{H}_5\text{F}^\ddagger$ zero-point level give the same product energy partitioning percentages. Relative translation receives $\sim 75\%$ of the product energy, with HF vibration the next highest recipient acquiring $\sim 10\%$.

The product energy distributions obtained from the classical and Wigner sampling are shown in Fig. 3. For each product energy, the distributions for the classical and Wigner sampling are in very good agreement. There are only two cases where the classical and Wigner bars of the distribution are not the same, given their statistical uncertainties. They are in the low energy region of the distributions for relative translation and C_2H_4 vibration. Nearly identical distributions are expected if a much larger ensemble of the direct dynamics trajectories was calculated.

B. $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier dynamics

1. Simulations with Wigner sampling

The trajectories with Wigner sampling of the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ zero-point level were performed in the same manner as trajectories with classical sampling of the zero-point level.⁵¹ The PES3 analytic potential energy function⁷² was used for the calculations. Direct dynamics simulations with MP2/6-31G* theory⁵⁶ give central barrier dynamics similar to what is found with this analytic function. The simulations with Wigner sampling were performed with a modified version of the VENUS computer program.^{67,68} The trajectories were integrated using a combined fourth-order Runge–Kutta and a sixth-order Adams–Moulton predictor–corrector algorithm.¹ A step size of 0.05 fs was used to obtain good energy conservation. Each trajectory was sent off the barrier in both forward and backward directions. The total integration time for a backward or forward trajectory is 20 ps. A reaction event is defined if the center-of-mass separation between Cl^- and CH_3Cl exceeds 30 Å within 20 ps for either the forward and backward integration.

The trajectories were initiated at the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier using the normal mode rigid rotor Hamiltonian. The reaction coordinate translational energy is 3.45 kcal/mol and rotational energy is sampled from a 300 K

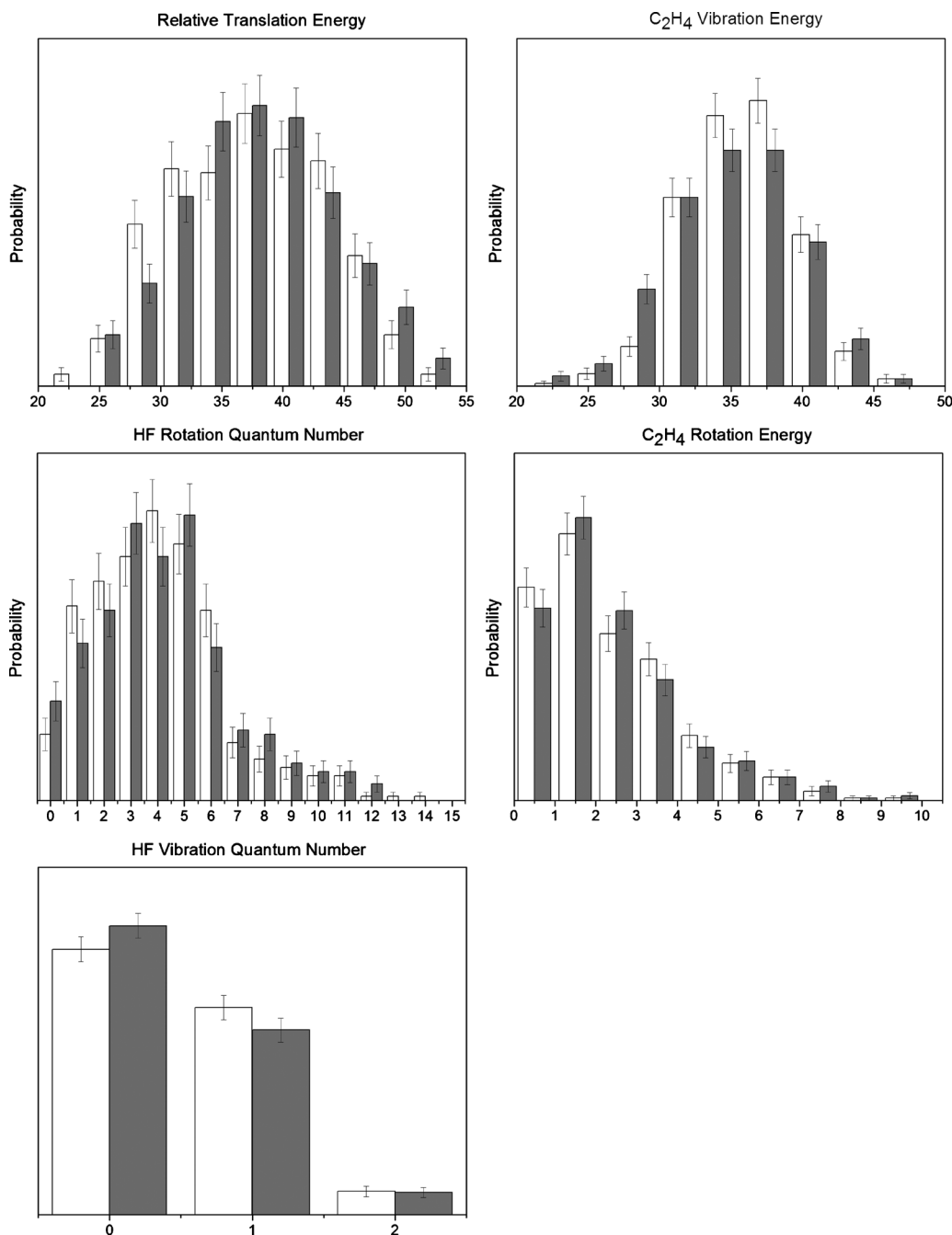


FIG. 3. Histograms of different distributions for $\text{C}_2\text{H}_5\text{F}^* \rightarrow \text{HF} + \text{C}_2\text{H}_4$ product energy partitioning. The white and gray bars are for classical and Wigner sampling of the TS zero-point energy level, respectively. The error bars in the histograms are the standard deviations. Energy is given in kcal/mol.

Boltzmann distribution. Zero-point energy is added to the $3N-7=11$ vibrational modes at the central barrier. A total of 800 trajectories was calculated for both the classical and Wigner sampling.

Values for the normal mode coordinates and momenta, chosen for the trajectory initial conditions using Wigner sampling, are in very good agreement with the Wigner distributions. The extent of agreement, though not shown, is similar to that for the $\text{C}_2\text{H}_5\text{F}^*$ sampling in Fig. 2.

2. Comparison of results for classical and Wigner sampling

As shown in the previous work, there are substantial non-TST (transition state theory) and non-RRKM (Rice–

Ramsperger–Kassel–Marcus) dynamics associated with the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier.^{51,56,64} In moving off the central barrier, the trajectories do not form an ion-dipole complex (either $\text{Cl}^-\cdots\text{CH}_3\text{Cl}$ for the backward direction or $\text{ClCH}_3\cdots\text{Cl}^-$ for the forward direction), which according to the RRKM theory should dissociate within the 20 ps integration time. Instead, a substantial number of barrier recrossings occur. For the ensemble of 800 trajectories calculated here, the number of nonstatistical barrier recrossings and the types of trajectories were analyzed. The forward trajectories generated by the Wigner sampling method have a total of 1359 nonstatistical barrier recrossings. In comparison, trajectories with classical sampling give 1329 nonstatistical barrier recrossings. Similarly, for the backward integrated trajectories,

TABLE II. Statistics of different types of events for trajectories initiated at the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier.

Sampling	Trajectory types ^a					
	$\text{C}_i \leftrightarrow \text{C}_i$	$\text{C}_i \leftrightarrow \text{C}_j$	$\text{R}_i \leftrightarrow \text{C}_i$ and $\text{C}_i \leftrightarrow \text{R}_i$	$\text{R}_i \leftrightarrow \text{C}_j$ and $\text{C}_i \leftrightarrow \text{R}_j$	$\text{R}_i \leftrightarrow \text{R}_i$	$\text{R}_i \leftrightarrow \text{R}_j$
Wigner	266	329	82	104	7	12
Classical	286	317	85	95	11	6

^aThe subscripts i and j denote different sides of the central barrier. C denotes either the $\text{Cl}^-\cdots\text{CH}_3\text{Cl}$ or $\text{ClCH}_3\cdots\text{Cl}^-$ ion-dipole complex, and R denotes one of the $\text{Cl}^- + \text{CH}_3\text{Cl}$ asymptotic limits. For example, $\text{C}_i \leftrightarrow \text{C}_i$ are events (i.e., trajectories) that start and end in the same ion-dipole complex, and $\text{R}_i \leftrightarrow \text{C}_j$ are events that start at one asymptotic limit and end in the ion-dipole complex on the other side of the central barrier.

there are 1489 and 1350 such recrossings, respectively. In total, there is only a 6% difference in the barrier recrossing numbers between the two types of sampling methods, an insignificant difference.

As described above, the trajectories are initiated at the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier and integrated in both the backward and forward directions. The combination of these two trajectories forms a “full” trajectory. Where the backward integrated “half” trajectory stops identifies the place the full trajectory starts. Where the forward integrated half trajectory stops identifies the place the full trajectory ends. In Table II, different types of full trajectories are listed and are compared for the two sampling methods. The trajectory types are identified in part by the sides of the potential energy barrier they start and end. The number of trajectories that start from one ion-dipole complex well and end in the same complex well (i.e., $\text{C}_i \leftrightarrow \text{C}_i$) is 266 for Wigner sampling and 286 for classical sampling. Trajectories that start from reactant (or product) on one side of the barrier and end in a complex on the same side of the barrier are denoted as $\text{R}_i \leftrightarrow \text{C}_i$ events. If this process is reversed, it is denoted as $\text{C}_i \leftrightarrow \text{R}_i$. Reactant (or product) and complex that start and end on different sides of the barrier are denoted as $\text{R}_i \leftrightarrow \text{C}_j$ or $\text{C}_i \leftrightarrow \text{R}_j$ events. The total number of $\text{R} \leftrightarrow \text{C}$ and $\text{C} \leftrightarrow \text{R}$ events, independent of i and j is 186 for Wigner sampling and 180 for classical sampling, which are nearly identical.

The numbers of $\text{R}_i \leftrightarrow \text{R}_i$ and $\text{R}_i \leftrightarrow \text{R}_j$ events are very small, i.e., they constitute only 2% of the trajectories, and there are differences between the numbers of these trajectory types for Wigner and classical sampling. However, they are not statistically significant. Wigner sampling gives twice as many $\text{R}_i \leftrightarrow \text{R}_j$ events as classical sampling does, but, if standard deviations are included, the numbers of these events for the two sampling methods are statistically the same.

Overall, for trajectories initiated at the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier, there are extensive nonstatistical barrier recrossings for both the classical and Wigner sampling methods. This nonstatistical barrier recrossing dynamics is in agreement with previous trajectory simulations.^{51,56} For these trajectories integrated for 20 ps in both the forward and backward directions off the central barrier, most started and ended in one of the ion-dipole complexes. This result is in agreement with the previous trajectory studies,^{51,56} but not with RRKM theory, which predicts the vast majority of the ion-dipole complexes should dissociate to $\text{Cl}^- + \text{CH}_3\text{Cl}$ within the 20 ps of the trajectory integrations. RRKM theory predicts that only a very small fraction of the ion-dipole complexes

should cross the central barrier and form the other ion-dipole complex. Thus, RRKM theory predicts negligible barrier recrossing.

IV. DISCUSSION

The trajectory simulations for these two reaction systems show that there are no significant differences in the results for Wigner and classical sampling of initial conditions at the reactions' transition states. Wigner and classical sampling of the trajectory initial conditions give statistically the same product energy partitioning for $\text{C}_2\text{H}_5\text{F}^\ddagger \rightarrow \text{HF} + \text{C}_2\text{H}_4$ dissociation and nonstatistical central barrier dynamics for $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$.

It is of interest to consider why Wigner and classical sampling of the TS's zero-point energy level give the same post-transition state dynamics for these two reactions. For Wigner sampling, which represents the quantum distribution in the classical phase space, the most probable value of a normal mode coordinate is its equilibrium value, while for classical sampling, the most probable values for a normal mode coordinate are its values at its classical turning points (see Fig. 1). Though this difference is striking, there are aspects of Wigner and classical sampling of a TS's vibrational energy level that are identical. For both Wigner and classical sampling, (1) the average value of a normal mode coordinate (or momentum) is zero; (2) the reaction coordinate is treated classically and placed at the saddle point with a specific translational energy; and (3) the $3N-7$ vibrational modes orthogonal to the reaction coordinate are sampled in accord with their vibrational quantum numbers n . Apparently, these features of the trajectory initial conditions are more important for determining the post-transition state dynamics than are the initial distributions of the TS's normal mode coordinates and momenta, and thus the initial phases between the normal mode coordinates and momenta.

Though Wigner and classical sampling give the same post-transition state dynamics for the two reactions studied here, for direct photodissociation, they often give substantially different dynamics. On the surface, the difference between post-transition state dynamics and direct photodissociation dynamics seems surprising since the potential energy release and motion from a TS to products is similar to the photodissociation dynamics on a repulsive PES. However, a more in depth analysis shows there are important differences in the samplings for post-transition state and photodissociation dynamics. For the latter, the sampling is performed for the $3N-6$ vibrational coordinates and momenta on the

ground state potential energy surface. This distribution is then projected onto the excited electronic state on which the photodissociation occurs. To compare with sampling at a TS, a reaction coordinate and $3N-7$ orthogonal vibration modes may be identified for the repulsive potential energy release of the excited state.⁷³ For the general situation, these degrees of freedom for the excited state will be linear combinations of the $3N-6$ normal modes of the ground electronic state. In addition, the equilibrium geometry for the $3N-7$ vibrational modes of the excited state is in general different than that for the ground state.

The above differences between the ground and the excited state PESs lead to significant differences between Wigner and classical sampling of initial conditions for the excited electronic state. These differences are not found for Wigner versus classical sampling at a TS. For the excited electronic state's initial conditions, Wigner and classical sampling will give different values for the reaction coordinate and its potential energy, and different values for the reaction coordinate's translational energy. These differences will clearly affect product energy partitioning for photodissociation. Since the equilibrium geometries differ for the vibrational modes of the ground and excited state PESs, it is expected that Wigner and classical sampling of the ground state PES will also give important differences in the initial conditions for the $3N-7$ vibrational modes of the excited electronic state. The average values of the coordinates for the excited state modes will not be zero and will differ for Wigner and classical sampling. Each vibrational mode of the excited electronic state will have a distribution of energies, and this distribution and its average value will differ for Wigner and classical sampling. Since the energies for the vibrational modes of the excited state project onto the product degrees of freedom, the different energies of the excited state's vibrational modes for Wigner and classical sampling will result in different product energies for these two sampling methods.

In summary, the studies presented here show that within statistical uncertainties, Wigner and classical sampling of TS vibrational energy levels give trajectories with the same post-transition state dynamics. Though Wigner and classical sampling give different distributions for the coordinates and momenta in phase space, they are similar in that they sample the same energy levels. For the two systems studied here, apparently this latter feature is more important than the former. In contrast, Wigner and classical sampling often give different photodissociation dynamics. Here the sampling is performed for a vibrational state of the ground electronic state PES, and this vibrational state is then projected onto the repulsive excited electronic state. Differences between the ground and the excited PESs result in different excitations of the vibrational and dissociative coordinates on the excited state PES following Wigner and classical sampling of the ground state's initial conditions. The result is different product distributions for the two sampling methods.

Finally, it is of interest to consider the possibility that for the two reactions studied here, the time scale for the motion from the TS to products may be long enough to couple the modes in such a way that irrespective of the initial sampling

the system recovers the classical random phase distribution. Thus, for much more direct post-transition state processes, such couplings may not take place and Wigner and classical sampling of the TS initial conditions may give different results. For the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ system, with its substantial barrier recrossings and trapping in the barrier and ion-dipole complex regions of the PES, the post-transition state dynamics is far from direct and the time scale is quite long. What is of interest is that Wigner and classical sampling give the same result even though the $[\text{Cl}\cdots\text{CH}_3\cdots\text{Cl}]^-$ central barrier dynamics is highly nonstatistical. $\text{C}_2\text{H}_5\text{F}^\ddagger \rightarrow \text{HF} + \text{C}_2\text{H}_4$ dissociation is a quite direct process, which partitions the product energy in a highly nonstatistical way and one in which Wigner and classical sampling give the same result. The question then is for even more direct processes do Wigner and classical sampling give different product energy partitioning distributions. This is an interesting question to address in future studies.

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- ¹D. L. Bunker, *Methods Comput. Phys.* **10**, 287 (1971).
- ²D. G. Truhlar and J. T. Muckerman, in *Atom-Molecular Collision Theory*, edited by R. B. Bernstein (Plenum, New York, 1976), pp. 505–566.
- ³G. H. Peslherbe, H. Wang, and W. L. Hase, *Adv. Chem. Phys.* **105**, 171 (1999).
- ⁴J. G. López, G. Vayner, U. Lourderaj, S. V. Addepalli, S. Kato, W. A. deJong, T. L. Windus, and W. L. Hase, *J. Am. Chem. Soc.* **129**, 9976 (2007).
- ⁵D. H. Lu and W. L. Hase, *J. Phys. Chem.* **92**, 3217 (1988).
- ⁶D. H. Lu and W. L. Hase, *J. Chem. Phys.* **91**, 7490 (1989).
- ⁷V. Bernshtein and I. Oref, *J. Chem. Phys.* **106**, 7080 (1997).
- ⁸W. A. Lester, Jr., in *Dynamics of Molecular Collisions, Part A*, edited by W. H. Miller (Plenum, New York, 1972), pp. 1–32.
- ⁹F. J. Aoiz, L. Bañares, and V. J. Herrero, in *Advances in Classical Trajectory Methods*, edited by W. L. Hase (JAI, London, 1998), Vol. 3, pp. 121–182.
- ¹⁰J. Mikosch, S. Trippel, C. Eichhorn, R. Otto, U. Lourderaj, J. X. Zhang, W. L. Hase, M. Weidemüller, and R. Wester, *Science* **319**, 183 (2008).
- ¹¹A. Untch, R. Schinke, R. Cotting, and J. R. Huber, *J. Chem. Phys.* **99**, 9553 (1993).
- ¹²R. Schinke, *Photodissociation Dynamics* (Cambridge University Press, New York, 1993), p. 93.
- ¹³S. R. Vande Linde and W. L. Hase, *J. Chem. Phys.* **93**, 7962 (1990).
- ¹⁴G. H. Peslherbe, H. B. Wang, and W. L. Hase, *J. Chem. Phys.* **102**, 5626 (1995).
- ¹⁵R. Wester, A. E. Bragg, A. V. Davis, and D. M. Neumark, *J. Chem. Phys.* **119**, 10032 (2003).
- ¹⁶E. Martínez-Núñez, A. Rahaman, and W. L. Hase, *J. Phys. Chem. C* **111**, 354 (2007).
- ¹⁷L. Yang, O. A. Mazzyar, U. Lourderaj, J. P. Wang, M. T. Rodgers, E. Martínez-Núñez, S. V. Addepalli, and W. L. Hase, *J. Phys. Chem. C* **112**, 9377 (2008).
- ¹⁸G. C. Schatz, in *Advances in Classical Trajectory Methods*, edited by W. L. Hase (JAI, London, 1998), Vol. 3, pp. 205–230.
- ¹⁹J. R. Schmidt, P. V. Parandekar, and J. C. Tully, *J. Chem. Phys.* **129**, 044104 (2008).

- ²⁰S. Y. Grebenshchikov, R. Schinke, and W. L. Hase, in *Unimolecular Kinetics. Part I. The Reaction Step*, edited by N. J. B. Green (Elsevier, New York, 2003), Vol. 39, pp. 105–242.
- ²¹R. Schinke, *Photodissociation Dynamics* (Ref. 12), p. 99.
- ²²E. Wigner, *Phys. Rev.* **40**, 749 (1932); *Z. Phys. Chem.* **19**, 203 (1933).
- ²³S. Goursaud, M. Sizun, and F. Fiquet-Fayard, *J. Chem. Phys.* **65**, 5453 (1976).
- ²⁴E. J. Heller, *J. Chem. Phys.* **65**, 1289 (1976).
- ²⁵A. García-Vela, *J. Chem. Phys.* **112**, 8302 (2000).
- ²⁶K. Husimi, *Proc. Phys. Math. Soc. Jpn.* **22**, 264 (1940).
- ²⁷A. García-Vela, J. Rubayo-Soneira, G. Delgado-Barrio, and P. Villarreal, *J. Chem. Phys.* **104**, 8405 (1996).
- ²⁸K. M. Christoffel and J. M. Bowman, *J. Chem. Phys.* **104**, 8348 (1996).
- ²⁹K. Takahashi, *Planet. Space Sci.* **98**, 109 (1989).
- ³⁰G. Delgado-Barrio, A. García-Vela, C. García-Rizo, M. I. Hernández, and P. Villarreal, *Mol. Eng.* **7**, 219 (1997).
- ³¹J. M. Bowman, A. Kupperman, and G. C. Schatz, *Chem. Phys. Lett.* **19**, 21 (1973).
- ³²E. J. Heller, *J. Chem. Phys.* **68**, 2066 (1978).
- ³³S. Goursaud, M. Sizun, and F. Fiquet-Fayard, *J. Chem. Phys.* **68**, 4310 (1978); M. Sizun and S. Goursaud, *J. Chem. Phys.* **71**, 4042 (1979).
- ³⁴C. Getino, B. Sumpter, and J. Santamaria, *Chem. Phys.* **145**, 1 (1990).
- ³⁵R. C. Brown and E. J. Heller, *J. Chem. Phys.* **75**, 186 (1981).
- ³⁶M. D. Pattengill, *Chem. Phys.* **68**, 73 (1982).
- ³⁷M. D. Pattengill, *Chem. Phys.* **75**, 59 (1983).
- ³⁸M. D. Pattengill, *Chem. Phys.* **78**, 229 (1983).
- ³⁹M. D. Pattengill, *Chem. Phys. Lett.* **105**, 651 (1984).
- ⁴⁰M. G. Sheppard and R. B. Walker, *J. Chem. Phys.* **78**, 7191 (1983).
- ⁴¹M. N. D. S. Cordeiro, E. Martínez-Núñez, A. Fernández-Ramos, and S. A. Vázquez, *Chem. Phys. Lett.* **381**, 37 (2003).
- ⁴²E. J. Heller and R. C. Brown, *J. Chem. Phys.* **75**, 1048 (1981).
- ⁴³H. W. Lee and M. O. Scully, *J. Chem. Phys.* **73**, 2238 (1980).
- ⁴⁴H. W. Lee and M. O. Scully, *J. Chem. Phys.* **77**, 4604 (1982).
- ⁴⁵R. Schinke, *J. Phys. Chem.* **92**, 3195 (1988).
- ⁴⁶A. García-Vela, R. B. Gerber, and J. J. Valentini, *J. Chem. Phys.* **97**, 3297 (1992).
- ⁴⁷A. Untch, S. Hennig, and R. Schinke, *Chem. Phys.* **126**, 181 (1988).
- ⁴⁸A. B. McCoy, Y. Hurwitz, and R. B. Gerber, *J. Phys. Chem.* **97**, 12516 (1993).
- ⁴⁹F. Chen and A. B. McCoy, *J. Phys. Chem. A* **112**, 9477 (2008).
- ⁵⁰U. Lourderaj, K. Park, and W. L. Hase, *Int. Rev. Phys. Chem.* **27**, 361 (2008).
- ⁵¹Y. J. Cho, S. R. Vande Linde, L. Zhu, and W. L. Hase, *J. Chem. Phys.* **96**, 8275 (1992).
- ⁵²C. Doubleday, Jr., K. Bolton, G. H. Peslherbe, and W. L. Hase, *J. Am. Chem. Soc.* **118**, 9922 (1996).
- ⁵³K. Bolton, W. L. Hase, and C. Doubleday, Jr., *Ber. Bunsenges. Phys. Chem.* **101**, 414 (1997).
- ⁵⁴E. Dong, D. W. Setser, W. L. Hase, and K. Song, *J. Phys. Chem. A* **110**, 1484 (2006).
- ⁵⁵L. Sun and W. L. Hase, *J. Chem. Phys.* **121**, 8831 (2004).
- ⁵⁶L. Sun, W. L. Hase, and K. Song, *J. Am. Chem. Soc.* **123**, 5753 (2001).
- ⁵⁷L. Sun, K. Park, K. Song, D. W. Setser, and W. L. Hase, *J. Chem. Phys.* **124**, 064313 (2006).
- ⁵⁸D. L. Bunker, *J. Chem. Phys.* **37**, 393 (1962).
- ⁵⁹D. L. Bunker and W. L. Hase, *J. Chem. Phys.* **59**, 4621 (1973).
- ⁶⁰N. Davison, *Statistical Mechanics* (McGraw-Hill, New York, 1962), p. 19.
- ⁶¹J. M. Hammersley and D. C. Handscomb, *Monte Carlo Methods* (Methuen, London, 1964), p. 36.
- ⁶²W. L. Hase and D. G. Buckowski, *Chem. Phys. Lett.* **74**, 284 (1980).
- ⁶³E. Arunan, S. J. Wategaonkar, and D. W. Setser, *J. Phys. Chem.* **95**, 1539 (1991).
- ⁶⁴W. L. Hase, *Science* **266**, 998 (1994).
- ⁶⁵L. Sun and W. L. Hase, *Rev. Comput. Chem.* **19**, 79 (2003).
- ⁶⁶W. L. Hase, K. Song, and M. Gordon, *Comput. Sci. Eng.* **5**, 36 (2003).
- ⁶⁷W. L. Hase, R. J. Duchovic, X. Hu, A. Komornicki, K. F. Lim, D.-h. Lu, G. H. Peslherbe, K. N. Swamy, S. R. Vande Linde, A. Varandas, H. Wang, and R. J. Wolf, *QCPE Bull.* **16**, 671 (1996).
- ⁶⁸X. Hu, W. L. Hase, and T. Pirraglia, *J. Comput. Chem.* **12**, 1014 (1991).
- ⁶⁹M. J. Frisch, G. W. Trucks, H. B. Schlegel *et al.*, GAUSSIAN 98, Gaussian, Inc., Pittsburgh, PA, 1998.
- ⁷⁰J. M. Millam, V. Bakken, W. Chen, W. L. Hase, and H. B. Schlegel, *J. Chem. Phys.* **111**, 3800 (1999).
- ⁷¹V. Bakken, J. M. Millam, and H. B. Schlegel, *J. Chem. Phys.* **111**, 8773 (1999).
- ⁷²D. J. Mann and W. L. Hase, *J. Phys. Chem. A* **102**, 6208 (1998).
- ⁷³K. Fukui, *Acc. Chem. Res.* **14**, 363 (1981).